

Creating a Prototype for a Data Anonymization Service

MASTER THESIS

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Declaration of Originality

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Erlangen, 30 July 2026

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Abstract

General instructions: If the thesis is written in German, this section contains the English translation. The abstract is 100–200 words maximum.

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Acronyms

DSGVO Datenschutz-Grundverordnung

GDPR General Data Protection Regulation

FLR functional legal requirements

FSR functional system requirements

FIR functional integration requirements

NFLR nonfunctional legal requirements

NFSR nonfunctional system requirements

NFIR nonfunctional integration requirements

LLM Large Language Model

UUID Universally Unique Identifier

BDSG Bundesdatenschutzgesetz (englisch Federal Data Protection Act)

ECJ European Court of Justice

1 Introduction

Deriving actionable insights from vast datasets is fundamentally driving innovation across diverse industries, from healthcare analytics to financial modeling. Yet, this heavy reliance on data accumulation presents a critical dichotomy. On one hand, organizations are legally and ethically mandated to protect individual privacy under frameworks such as the General Data Protection Regulation (GDPR) and emerging global standards. On the other, they remain wholly dependent on utilizing that same data to train machine learning models and perform complex statistical analyses.

In the specialized field of software engineering, this conflict is centered on the vast amount of metadata generated by development teams through their daily interactions with tools like GitHub and Jira. Effectively harnessing this information to drive productivity while complying with privacy laws requires innovative frameworks that can navigate the complexities of high-dimensional data.

The MECOIS project, a research initiative from the University of Erlangen, addresses the "Big Data" challenge within the specific domain of software engineering by transforming the vast metadata generated during development into actionable intelligence. Software development teams leave a continuous trail of activity-based artifacts in repositories such as GitHub and Jira, creating a high-dimensional data set that is traditionally difficult to interpret due to overlapping metrics, complexity, and incompatible value ranges. MECOIS solves these issues by applying statistical methods—specifically the creation of a composite index—to group correlating metrics into key dimensions. This data-driven approach provides an objective Productivity Index that allows organizations to compare performance across teams and projects, identifying early-stage bottlenecks like miscommunication or resource gaps while uncovering the best practices of high-performing teams. (Riehle, 2025)

To address the tension between data utilization and privacy, this thesis proposes a prototype for a dedicated anonymization service designed to process the metadata collected within the MECOIS framework. While MECOIS extracts high-dimensional, activity-based artifacts from platforms such as GitHub and Jira to

1. Introduction

provide deep insights into team productivity, the granular nature of this data — including timestamps, commit details, and developer identities — presents significant privacy risks. The proposed service aims to anonymize this data in strict accordance with the GDPR, ensuring that the trail of artifacts used to calculate the MECOIS Productivity Index is stripped of personally identifiable information. By integrating this anonymization layer, the project seeks to enable the MECOIS's mission of empowering software developers through data-driven analysis while maintaining a privacy-first approach that meets modern legal and ethical standards.

2 Key Terms and Definitions

2.1 German Privacy Laws

In Germany data privacy is regulated in two laws: The Datenschutz-Grundverordnung (DSGVO) and the Bundesdatenschutzgesetz (englisch Federal Data Protection Act) (BDSG).

2.1.1 General Data Protection Regulation

The DSGVO (often also abbreviated to DS-GVO) is the German name for the GDPR, which is a European Union regulation designed to ensure the protection of personal data and the free flow of data in Europe. It has been mandatory in all EU member states since May 25, 2018, and applies equally to private companies, public authorities, associations, and individuals. The GDPR strengthens the rights of EU citizens, as they now enjoy the same legal certainty in all member states and can decide how their own data is processed. For companies, it simplifies cross-border activities through uniform rules of conduct and the so-called “one-stop-shop” procedure, but also provides for substantial fines in the event of violations. While it is directly applicable in all EU member states, individual countries still have their own national data protection laws to supplement it. (für den Datenschutz und die Informationsfreiheit [BfDI], 2025)

2.1.2 Federal Data Protection Act

Germany passed its own privacy laws in 1977 called the . It had to be revised upon the adoption of the GDPR. Today the BDSG and DSGVO complement each other, since the GDPR has escape clauses, where the BDSG steps in. This applies, for example, to the processing of employee data or the establishment of supervisory authorities. (BfDI, 2025) (Däubler, 2017, pp. 68–70)

2.2 Pseudonymization and Anonymization

To comply with the data privacy laws, data often has to be pseudonymized or anonymized.

2.3 Pseudonymization

According to Article 4 No. 5 of the GDPR, pseudonymization is defined as the processing of personal data in such a manner that it can no longer be attributed to a specific data subject without the use of additional information. This additional information, often referred to as a "key" for re-identification, must be kept separately and be subject to rigorous technical and organizational measures to ensure that the personal data is not assigned to an identified or identifiable natural person. Crucially, pseudonymization is characterized as a processing operation rather than a static state, involving the active transformation of cleartext data into pseudonyms to mitigate risk while maintaining the utility of the data set. Because the underlying connection to a natural person is merely obscured rather than permanently severed, pseudonymized data remains classified as "personal data" and thus stays fully within the legal scope and obligations of the GDPR. (Members of the "Deutsche Gesellschaft für Medizinische Informatik et al. [GMDS], 2024, pp. 8, 21, 71) (Schwartzmann et al., 2022, p. 14)

2.4 Anonymization

In contrast, anonymization is the process of transforming personal data into anonymous information so that the data subject is no longer identifiable. While not explicitly defined within the text of the GDPR itself, it is recognized as a process that results in data that does not relate to an identified or identifiable natural person. A successful anonymization is fundamentally irreversible; once the process is complete, a re-identification of the individual must be impossible for the controller or any third party. Within the academic and legal discourse, a distinction is often made between "absolute anonymization," where re-identification is factually impossible for everyone, and "factual" or "relative anonymization," where re-identification would require a disproportionate effort in terms of time, cost, and labor. Once data is effectively anonymized, it is no longer considered personal data, and the requirements of the GDPR cease to apply. (GMDS, b., BvD, 2024, pp. 1, 9–10, 13, 21, 68, 73) (Schwartzmann et al., 2022, pp. 3, 16–19, 23)

2.5 Difference Pseudonymization and Anonymization

The fundamental difference between the two terms lies in the reversibility of the process and the subsequent legal classification of the data. Pseudonymization is a reversible procedure where the link between the data and the individual is preserved via a separate key, meaning the data retains its character as "personal data". Anonymization, however, aims for an irreversible state that permanently breaks this link, thereby removing the data from the regulatory framework of the GDPR. Furthermore, pseudonymization requires the controller to implement strict silos and access controls to manage re-identification information, whereas anonymization requires the permanent deletion or sufficient generalization of all direct and indirect identifiers to ensure that no such "key" exists to restore the original identity. (GMDS, b., BvD, 2024, pp. 8, 13, 31, 68) (Schwartmann et al., 2022, pp. 3, 19)

2. Key Terms and Definitions

3 Literature Review

This section reviews existing scientific publications to establish a baseline for data anonymization services within software engineering productivity metrics.

3.1 Data Protection Laws

While extensive literature exists on the GDPR and data anonymization, existing studies typically address these concepts at a high level or focus primarily on surveillance applications. Consequently, domain-specific literature directly aligned with the scope of this research remains scarce.

The standard reference work on employee data protection is Däubler (2017). Written in a practical style and accessible even to non-specialists, this handbook explains the entire field of data protection law. It focuses on employee data protection and the participation rights of works councils and staff councils. It also addresses the challenges posed by AI that are already apparent today, such as those arising in the hiring process or when working with ChatGPT. Another key focus is on the diverse means available to ensure that data protection is effectively enforced in practice. These range from data backup and the activities of the company data protection officer and the supervisory authority to the imposition of substantial fines and numerous claims for damages. The now very extensive case law of the European Court of Justice (ECJ) — which has set many new precedents — is analyzed in relation to these and numerous other questions. As a rule, this case law strengthens the protection of individuals, which is also of great significance for the scope of action available to works councils and staff councils. This shift has not yet been fully recognized everywhere.

In the context of data protection and research, the operationalization of de-identification procedures under the DSGVO is addressed by two central German guidelines that provide both technical and legal frameworks. Schwartzmann et al. (2022) establishes a structured procedural model that distinguishes between theoretical "absolute anonymity" and the more practically relevant "factual anonymization," where re-identification is considered impossible due to the

disproportionate effort required in terms of time, cost, and manpower. Complementing this, GMDs, b., BvD (2024) focuses specifically on the sensitive domain of health data under Art. 9 GDPR, highlighting the necessity of aligning national definitions with the broader European legal framework provided by Directive (EU) 2019/1024. While Schwartzmann et al. (2022) categorizes technical methods—such as randomization (e.g., differential privacy) and generalization (e.g., k-anonymity)—into specific application classes like AI algorithm training or software testing, GMDs, b., BvD (2024) provides detailed risk assessments for specific attack scenarios, including “singling out,” inference, and linkage attacks. Both sources emphasize that anonymity is not a static state but a dynamic process that must be continuously evaluated against the evolving “state of science and technology”. Furthermore, they converge on the legal requirement that both anonymization and pseudonymization constitute “processing” activities under the GDPR, therefore requiring a valid legal basis and remaining subject to the controller’s accountability obligations.

3.2 Server Design

TODO

3.3 Identity Management

The MECOIS project uses the tool SortingHat from the tool set GrimoireLab by CHAOSS for managing identities.

Dueñas et al. (2021) present GrimoireLab, an integrated and modular open-source tool set developed to facilitate the comprehensive retrieval, storage, and visualization of data from software repositories. This framework addresses a critical gap in software analytics by providing automated support for approximately 30 different data sources, including version control systems like Git, issue trackers such as Jira and Bugzilla, and communication platforms like Slack and mailing lists. Central to its architecture is a multi-stage pipeline that employs specialized components: Perceval for uniform data retrieval via JSON documents, Graal for detailed source code analysis, and SortingHat for advanced identity management and cross-platform contributor merging. Furthermore, GrimoireLab utilizes a distinct storage model that maintains “raw” copies of original data alongside “enriched” indexes optimized for visualization, a strategy that significantly enhances data maintainability, traceability, and research reproducibility. By providing a mature, field-tested framework for both academic research and industrial application, the authors demonstrate that GrimoireLab reduces the manual effort required for empirical software engineering studies while enabling the longitudinal tracking of developer activity and project health.

SortingHat is the specialized identity management component within the GrimoireLab framework, specifically designed to address the challenge of contributor fragmentation across diverse software repositories. Because developers frequently utilize different usernames, aliases, and email addresses across platforms such as Git, Slack, and Jira, SortingHat provides the essential infrastructure to unify these disparate identities into unique, individual profiles. The component operates by maintaining a dedicated relational database that stores identity data, affiliation tags, and the original source of each identity to ensure research traceability. To maintain high data quality, SortingHat utilizes a conservative algorithmic approach to merging—avoiding false positives caused by shared generic accounts—which is further complemented by HatStall, a web-based interface that allows researchers to manually curate identities and manage organizational affiliations. This process of identity unification is fundamental to the broader GrimoireLab pipeline; it allows the analytics engine to enrich data points with a unique contributor Universally Unique Identifier (UUID), thereby enabling the longitudinal tracking of developer trajectories and cross-platform activity metrics that would otherwise remain siloed. (Dueñas et al., 2021)

3. Literature Review

4 Requirements

The purpose of this chapter is to define the requirements necessary to facilitate a GDPR-compliant productivity benchmarking system. ‘ISO/IEC/IEEE International Standard - Systems and software engineering–Vocabulary’ (2017) splits requirements into functional and nonfunctional ones. Functional requirements define what a product must do and nonfunctional ones declare how the system should operate. Furthermore, the requirements can be classified into legality, system and integration. The service must abide to relevant data protection laws. System requirements specify how the service itself will be designed according to the author. The MECOIS project itself sets some integration demands in its contributing guidelines (‘MECOIS Github Repository’, 2026), that must be satisfied.

4.1 Functional Requirements

The table 4.1 specifies what the anonymization must do. They are sorted by functional legal requirements (FLR), functional system requirements (FSR) and functional integration requirements (FIR).

The Data Anonymization requirement FLR-1 is defined very vague by design. There can be many ways to anonymize data according to the DSGVO. Chapter 6 will compare and present different approaches. It stands in direct conflict with FSR-1 Preserve Data. The aim is to find a compromise between these two.

4.2 Nonfunctional Requirements

The table 4.2 specifies how the anonymization service must operate. They are sorted by nonfunctional system requirements (NFSR) and nonfunctional integration requirements (NFIR). There are no nonfunctional legal requirements (NFLR), because it is evaluated in chapter 6, how the legal requirements will be satisfied.

Table 4.1: Functional requirements

ID	Requirement	Description
FLR-1	Data Anonymization	The service must follow the GDPR which is defined by the DSGVO in Germany.
FSR-1	Preserve Data	The service must preserve all available Data.
FSR-2	independent Deployment	The service can be deployed and run independent from the rest of the MECOIS project.
FIR-1	Seamless Integration	The service must integrate into the project with as few changes as possible to existing features.

Table 4.2: Nonfunctional requirements

ID	Requirement	Description
NFSR-1	Containerized Service	The service must run in a docker container.
NFIR-1	Programming Language	The Backend of the MECOIS project is written in python3. Therefore, this project must also use python3. ('MECOIS Github Repository', 2026)
NFIR-2	Licence Policy	The contributing guidelines ('MECOIS Github Repository', 2026) state: "You are allowed to use any package that can help you do your work, as long as it is not a copyleft or proprietary license. However, it is still a good idea to ask beforehand, so you do not put in unnecessary work."
NFIR-3	Styleguide and Clean Code	The contributing guidelines ('MECOIS Github Repository', 2026) state: "We follow the PEP8 standard (with minor exceptions like allowing longer line limits). You can run the custom linter using: pylint. Additionally, your code must follow these rules before being merged into the main branch: • Remove comments that are not necessary for understanding your code • Remove unnecessary print statements • Make sure to update and provide the DocString description of the methods you write. Make sure to have it adjusted to your use-case if you copy method-stubs from other part of the code"

4. Requirements

5 Architecture

The MECOIS setup consists of 3 separate systems, that communicate via HTTP (see Figure 5.1). The systems are separated to protect sensitive information.

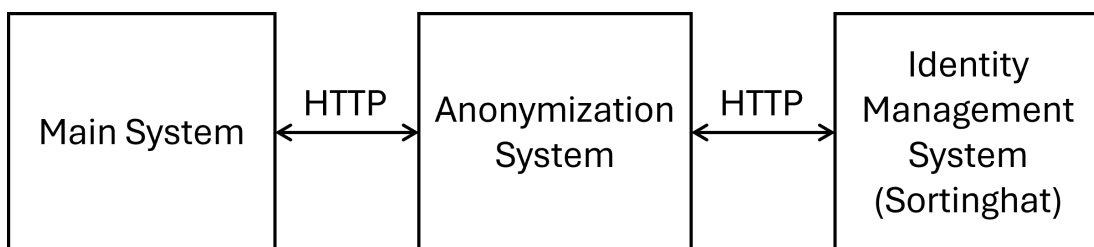


Figure 5.1: Basic System Architecture

5.1 Main System

The main systems is responsible everything but the identity management and therefore has many components.

5.1.1 Components

Figure 5.2 displays an overview of all components. (‘MECOIS Github Repository’, 2026)

Sources

The sources define the origin for the data. Currently we are using Git (GitHub and GitLab), Confluence and Jira as source. For Example from GitHub we pull among other commits, issues and pipelines.

Infrastructure

The infrastructure provides the database for the project. The data is stored in a data lake, which is a repository, where data is stored as-is without further

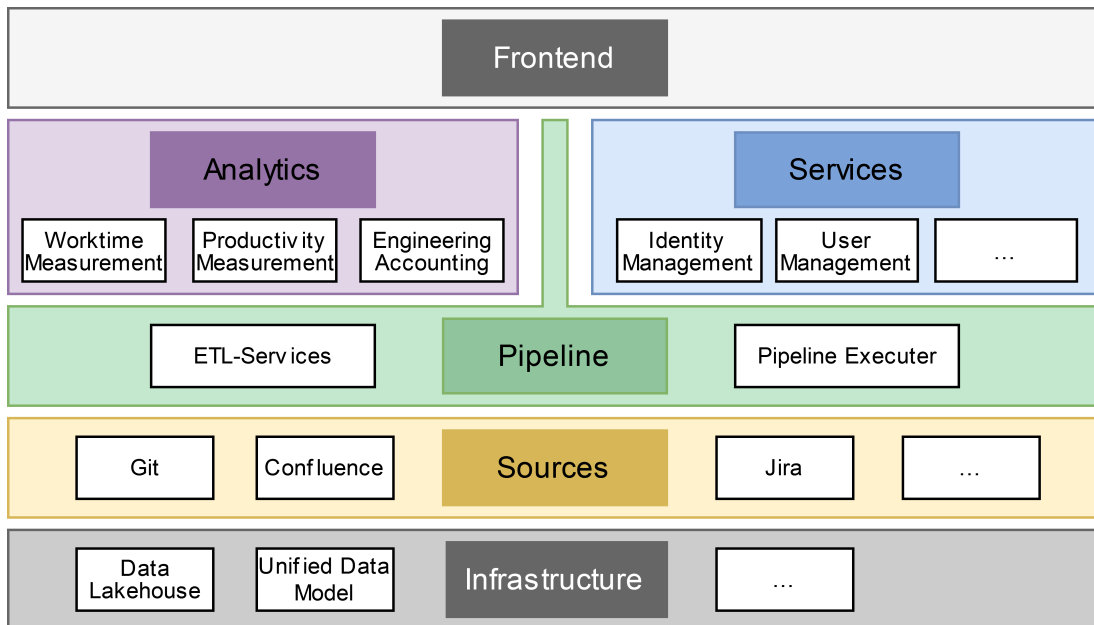


Figure 5.2: Main System Components (‘MECOIS Github Repository’, 2026)

structure (‘What is a data lake?’, 2026). Each set of data passes through 4 Stages: raw, bronze, silver, gold.

Pipeline

The pipelines transform the data between the stages.

Analytics

In the last step the data is used to provide multiple analytics for customers. Examples for such analytics are:

- Calculating the Costs of Inner Source Collaboration by Computing the Time Worked (Buchner & Riehle, 2021)
- Estimating Software Contribution Time from Git and GitHub Metadata (Stenzel, 2026)

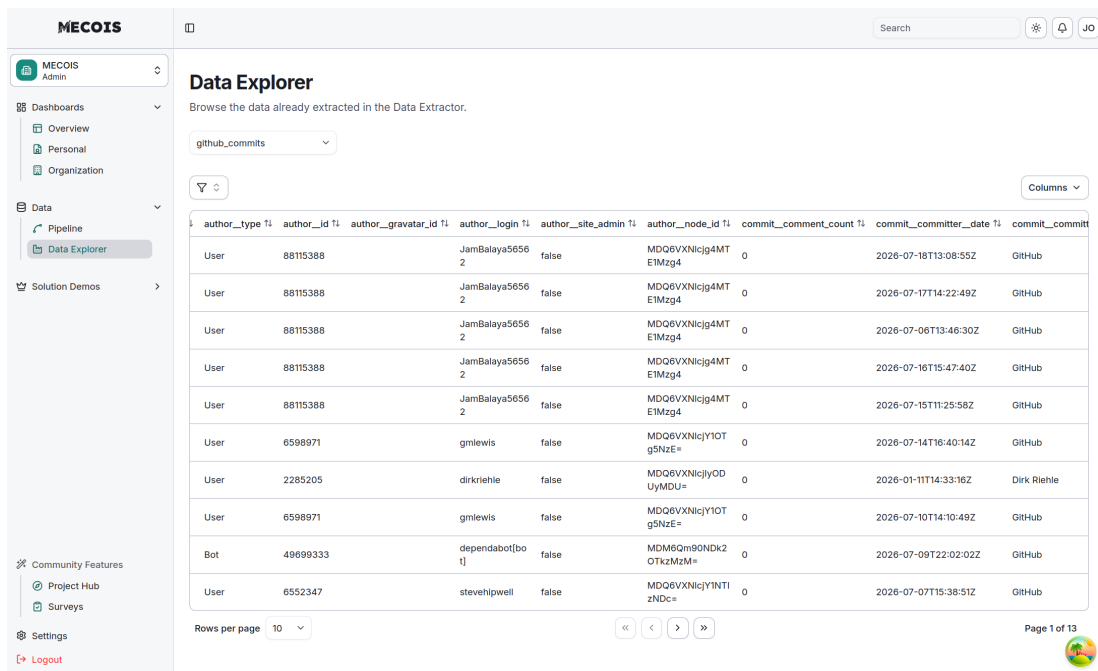
Services

Services are helper modules for pipelines and analytics. Such helper modules are:

- AI: a framework for communicating with a Large Language Model (LLM).
- config handling: Parses configuration files for pipelines.

Front end

The front end provides a Dashboard for the user-facing components of MECOIS. Figure 5.3 shows the Data Explorer tool, where users can examine the data in the data lake. Figure 5.4 presents the Data Extractor tool, that can be used to query data from the sources and feed it to the pipelines.



author_type	author_id	author_gravatar_id	author_login	author_site_admin	author_node_id	commit_comment_count	commit_committer_date	commit_committ
User	88115388	JamBalaya56562	false	MDQ6VXNjcjg4MT	0	2026-07-18T13:08:55Z	GitHub	
User	88115388	JamBalaya56562	false	MDQ6VXNjcjg4MT	0	2026-07-17T14:22:49Z	GitHub	
User	88115388	JamBalaya56562	false	MDQ6VXNjcjg4MT	0	2026-07-06T13:46:30Z	GitHub	
User	88115388	JamBalaya56562	false	MDQ6VXNjcjg4MT	0	2026-07-16T15:47:40Z	GitHub	
User	88115388	JamBalaya56562	false	MDQ6VXNjcjg4MT	0	2026-07-15T11:25:58Z	GitHub	
User	6598971	gmilewis	false	MDQ6VXNjcjY1OT	0	2026-07-14T16:40:14Z	GitHub	
User	2285205	dirkriehle	false	MDQ6VXNjcjY1OD	0	2026-01-11T14:33:16Z	Dirk Riehle	
User	6598971	gmilewis	false	MDQ6VXNjcjY1OT	0	2026-07-10T14:10:49Z	GitHub	
Bot	49699333	dependabot[bot]	false	MDM6Mm90NDk2	0	2026-07-09T22:02:02Z	GitHub	
User	6552347	stevhipwell	false	MDQ6VXNjcjY1NTI	0	2026-07-07T15:38:51Z	GitHub	

Figure 5.3: MECOIS Frontend Data Explorer

Orchestrators

The orchestrators can be used to run the pipeline. The set up the configuration and run the pipeline for the selected sources.

5.1.2 Data Flow

After pulling the source the raw data is stored exactly as provided as JSON. In the data lake the Data is processed multiple times and the states are called bronze, silver and gold. The processing is done by the pipelines.

Each pipeline has 3 steps:

1. Read the data from the data lake
2. transform data
3. write the transformed data to the data lake

5. Architecture

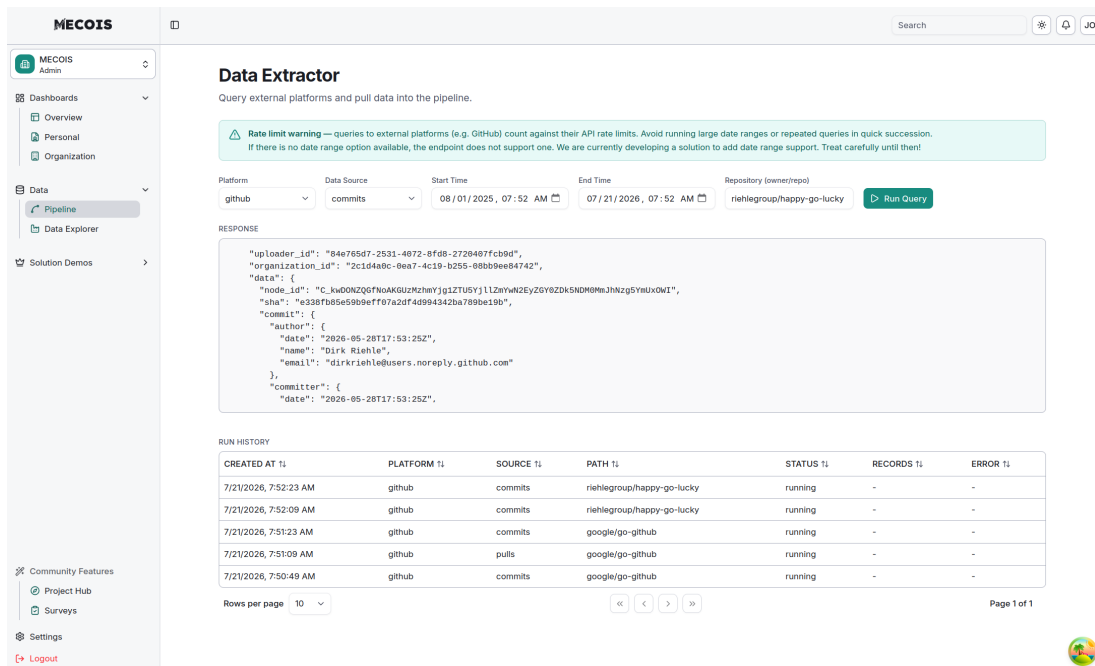


Figure 5.4: MECOIS Frontend Data Extractor

Figure 5.5 displays an overview of all components. (‘MECOIS Github Repository’, 2026)

JSON to Bronze

The `JSONtoBronzePipeline` processes the raw JSON data provided by the sources and writes it unchanged to the Data lake. The data itself isn’t changed in any way. Each source has an own table.

Bronze to Silver

The `BronzetoSilverPipeline` processes the bronze data in multiple steps to be ready for the analytics.

Flatten table The bronze state cannot represent nested data. (e.g. a GitHub commit contains an author and the author itself has multiple properties). In the first step those nested data in columns is split in multiple columns, so that each on only contains a single data point.

Drop irrelevant columns Not all data is useful. Some columns (e.g. the URL to the avatar of a GitHub user) are dropped.

Fix dates Dates are still stored as String. This steps casts these Strings into

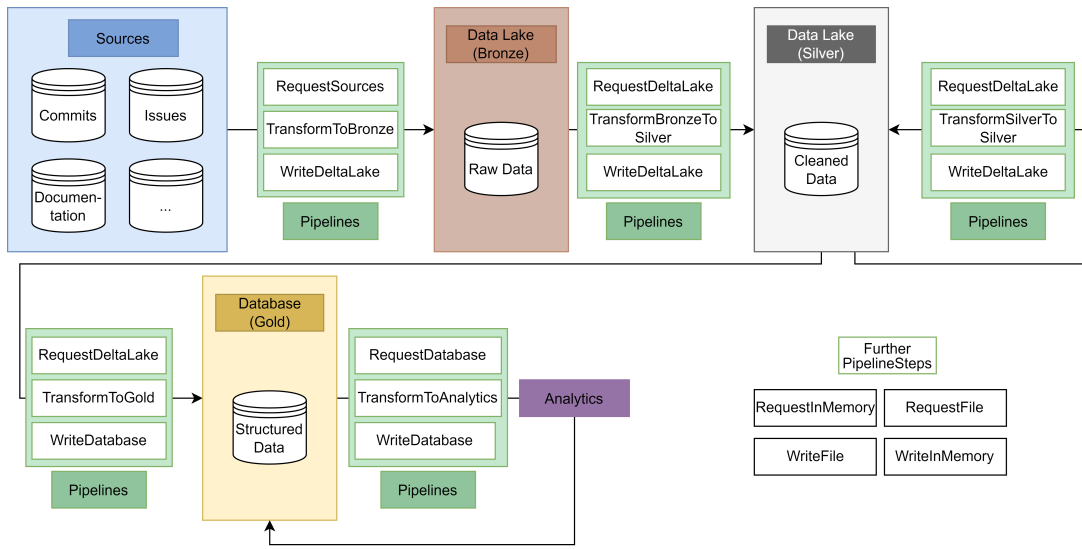


Figure 5.5: Data flow (‘MECOIS Github Repository’, 2026)

Anonymize Data The data is sent to the anonymization service for processing.

add UUID Each table has a way to create an UUID by concatenating some columns and hashing the result.

Silver to Silver

Identities may update in the identity management system. The `UpdateIdentityPipeline` updates any identities in the data lake that have been updated in the SortingHat.

Silver to Gold

There is no single `Silver2GoldPipeline`. The analytics represent the gold to silver pipelines. Each analytic uses the silver data to provide some metric.

Gold to Gold

Some analytics may need some mechanism to update the generated metric. This process is defined as gold to gold transformation.

5.2 Anonymization System

The anonymization system provides the anonymization service, that anonymizes the data according to the GDPR. It provides a client library for sending data and requesting identities.

5.3 Identity Management System

The identity management system is responsible for storing all information about individuals. The project is using the tool SortingHat from the tool set Grimoire-Lab by CHAOSS. This application is provided under the GPL-3.0 license.

One individual may have multiple identities on different platforms. The system can pool these different accounts into one individual. These individuals can also be grouped into teams. Multiple teams can form an organization. ('SortingHat', 2026) ('SortingHat Github Repository', 2026)

5.3.1 Identities

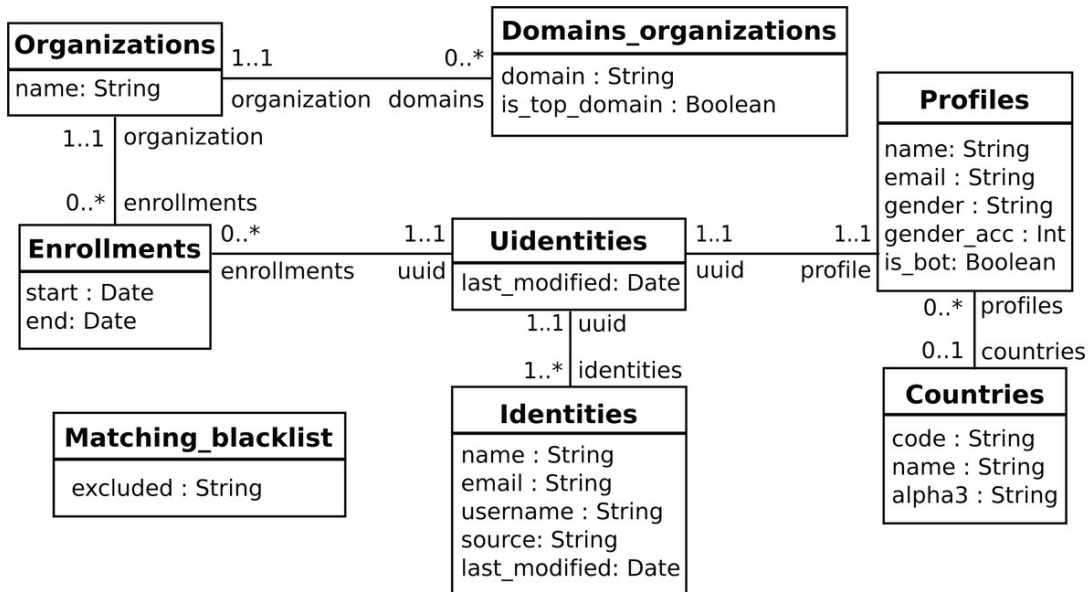


Figure 5.6: SortingHat UML diagram (Dueñas et al., 2021, p. 15)

Figure 5.6 describes how identities are managed in the SortingHat. One individual may have multiple identities on different platforms. The system can pool these different accounts into one individual. One individual has a UUID and is linked to a profile containing personal information. These individuals can also be grouped into organizations. ('SortingHat', 2026) ('SortingHat Github Repository', 2026)

5.3.2 Front end Dashboard

The SortingHat provides a dashboard, where individuals can be merged and associated into teams and organizations.

Figure 5.7 shows a screenshot of the front end. Since the screenshot contains real data, the individuals are blacked out. The individuals section can be seen in figure 5.8.

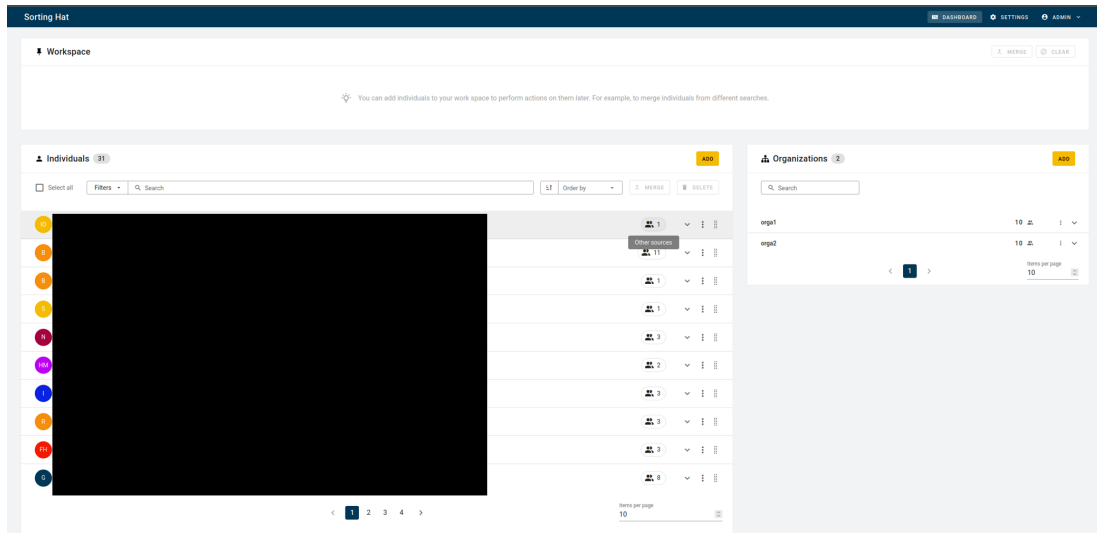


Figure 5.7: SortingHat front end (individuals are censored for privacy reasons)

5.3.3 API

The SortingHat also provided an API for automating interactions, like listing, updating and deleting individuals.

5. Architecture

Individuals

9

ADD

Select all

Filters

Search

Name

MERGE

DELETE

EM

Eva Millán

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Figure 5.8: SortingHat individuals section (uncensored) (‘SortingHat’s new interface’, 2026)

6 Design and Implementation

Chapter 5 described the architecture of the MECOIS project and the integration of the anonymization service. This chapter will explain how the service is implemented.

6.1 Reason for an Anonymization Service

According to the BDSG, personal data may only be collected, processed, or used for a specific purpose. It is forbidden to use this data for other purposes. There are only a few specific Exceptions for this rule. (Däubler, 2017, p. 72)

6.1.1 Purpose Limitation

Since software development tools like GitHub, GitLab or Jira contains personal data (e.g. git commit author name or Jira ticket reporter), its use falls under the GDPR. In most cases there is no written agreement on how the data can be used, so it can be assumed that the purpose for this data is to help developers organize their work. Generating productivity measurements for individual employees would be considered a misuse of the data. Däubler (2017, pp. 266–267) explicitly states that data that consists of operational workflows gathered by electronic systems cannot be used for measuring productivity of individual employees. (Däubler, 2017, pp. 266–267)

One exception is scientific research. If individuals are part of a scientific research project, their data can be used for this purpose. The data should get pseudonymized or anonymized as fast as possible, in order to minimize the infringement of privacy of the individual members.(Däubler, 2017, pp. 273–274)

6.1.2 Anonymized Data

The BDSG only applies to personal Data, which is defined in Section 3 (1): Personal data consists of specific information about the personal or factual circumstances of an identified or identifiable natural person (data subject). (Däubler,

2017, p. 69)

The DSGVO has a similar approach. In Article 4(1) it defines personal data as any information relating to an identified or identifiable natural person (hereinafter referred to as the “data subject”). A natural person is considered identifiable if they can be identified, directly or indirectly, in particular by association with an identifier such as a name, an identification number, location data, an online identifier, or one or more factors specific to the physical, physiological, genetic, mental, economic, cultural, or social identity of that natural person. (Däubler, 2017, p. 69)

Unlike anonymized data, where re-identification is impossible, pseudonymized data is still traceable to the original person. Therefore pseudonymized data has to comply with the DSGVO and BDSG. Anonymized data is not mentioned in the DSGVO because, since it does not relate to specific individuals, it is not subject to the DSGVO. (Däubler, 2017, p. 69)

6.1.3 Consequences for MECOIS

For the MECOIS project the exception for scientific research is very important regarding the development of the project. As long as MECOIS is a scientific research project, the data can be used to generate productivity measurements as long as the process serves scientific research purposes. But the data still should be pseudonymized or anonymized.

Regarding the final product, MECOIS must anonymize the Data. Otherwise it is illegal to produce productivity metrics.

6.2 Identifying Identifiers

The identification and categorization of identifiers is a fundamental prerequisite for implementing effective data protection measures, such as pseudonymization or anonymization. Attributes within a data set are typically classified into three categories: direct identifiers, indirect (or quasi-) identifiers, and sensitive attributes. (Krebs & Hagenweiler, 2022, pp. 87–88, 131–132) (Schwartmann et al., 2022, pp. 9–10) (GMDS, b., BvD, 2024, p. 37)

Direct Identifiers Direct identifiers, such as civil names, social security numbers, or email addresses, allow for the immediate identification of a natural person without further processing. (Krebs & Hagenweiler, 2022, pp. 87–88) (GMDS, b., BvD, 2024, pp. 37, 71)

Indirect Identifiers Indirect identifiers include data points like birth dates, ZIP codes, or rare job titles that do not identify a person in isolation but can lead to identification when combined with external datasets or background knowledge — a risk often termed "singling out". (Krebs & Hagenweiler, 2022, pp. 87–88) (Schwartmann et al., 2022, pp. 5, 13) (GMDS, b., BvD, 2024, p. 37)

Sensitive Attributes Sensitive attributes, such as the special categories of data protected under Article 9 of the GDPR (e.g., genetic or health data), require the highest level of protection due to the potential for discrimination and significant impact on individual freedoms. (Krebs & Hagenweiler, 2022, pp. 131–133) (GMDS, b., BvD, 2024, pp. 8, 24–25)

Ultimately, the classification of any specific data point is context-dependent, as even non-identifying data may become identifying depending on the "reasonable" means and background information available to a potential attacker. (GMDS, b., BvD, 2024, pp. 36–37, 54)

Example GitHub Commit In the context of GitHub commits, direct identifiers are: names and emails of author and committer. Since every employee has access to the GitHub repositories, indirect identifiers like the commit message (including the timestamp), ids and specific URLs can be easily resolved to the involved persons. Sensitive attributes are not present in GitHub.

6.3 Approaches for Pseudonymization and Anonymization

Techniques for pseudonymization and anonymization are categorized based on their technical approach and the type of data they address.

6.3.1 Pseudonymization Techniques

Pseudonymization involves replacing identifiers with surrogates so that the data cannot be attributed to a specific person without additional information stored separately.

Masking and Replacement Direct identifiers are replaced with constant values, characters, or placeholders (e.g., replacing "Max Muster" with "Person A").

Shuffling (Mixing) Values within a column are swapped or rearranged across different records through random distribution to remove the link between attributes and specific individuals.

Variance Method For quantitative data, numerical values are altered by adding random deviations within a specified range while maintaining overall statistical distribution.

Cryptographic Encryption Identifiers are transformed using a secret key, making the process reversible for authorized parties who hold the key.

Hashing A one-way mathematical function maps identifiers to a fixed-length string (hash value); techniques like adding Salt or Pepper (random characters) are used to prevent brute-force or rainbow table attacks.

Tokenization Often used in the financial sector, this replaces sensitive identifiers like card IDs with randomly generated tokens.

6.3.2 Anonymization Techniques for Structured Data

Anonymization aims to permanently remove the connection between data and a natural person, making re-identification practically impossible.

Randomization (Perturbation)

These methods distort attribute values to decrease the probability of identifying individuals from a dataset.

Stochastic Overlay Merkmale are altered with random variables so that the specific values are less accurate but general distribution is preserved.

Swapping Attribute values are shifted between different data records, which is particularly effective when combined with the removal of quasi-identifiers.

Falsification (Noise Addition) Artificial statistical noise is added to some or all data points to prevent reliable estimation of an individual's original values.

Microaggregation (Clustering) Data records are grouped by similarity, and their values are replaced by a representative average (e.g., mean or median) for that group.

Generalization

These non-perturbative methods reduce the granularity of the data to hide individuals within groups.

Aggregation Summarizing individual values into larger units or categories.

Recoding Grouping values into intervals (e.g., changing exact ages to age ranges like "20-30") or higher-level categories.

Suppression Specific values, cells, or entire columns are deleted or replaced with a special suppression value to eliminate outliers or highly identifying information.

Anonymity Models and Criteria

These models define thresholds for the degree of protection achieved after applying technical methods.

k-Anonymity Ensures that every individual record is indistinguishable from at least $k-1$ other records based on its quasi-identifiers.

l-Diversity Extends k-Anonymity by ensuring there is sufficient variety in the sensitive attributes within each group of indistinguishable records to prevent attribute disclosure.

t-Closeness Refines l-Diversity by requiring that the distribution of a sensitive attribute in any group is close to the distribution of that attribute in the overall population.

-Presence Limits the probability that an attacker can determine whether a specific person is present in a private dataset by comparing it with a publicly available one.

Advanced and KI-Based Methods

Differential Privacy A mathematical framework that adds controlled statistical noise to query results or base data, ensuring that the presence or absence of a single individual does not significantly affect the outcome.

Data Synthesis Statistical models or Generative Adversarial Networks (GANs) are used to create artificial datasets that reflect the properties of the original data without including real personal identifiers.

Semantic Anonymization Uses active ontologies and inferencing to transform data semantically while preserving its analytical value.

Federated Machine Learning AI models are trained locally on decentralized data sources, and only non-personal model parameters are transmitted to a central coordinator.

Techniques for Unstructured Data

Metadata Removal Deleting fields in file formats like PDF or DICOM (medical imaging) that contain authors, dates, or study IDs.

Natural Language Processing (NLP) Automatically identifying and masking entities like names, locations, or companies in free text.

Paraphrasing Replacing identifying phrases with generic descriptions while preserving the document's readability.

Author Obfuscation Masking an author's unique writing style using neural translation methods like A4NT.

Multimedia Processing Techniques for images and video include pixelation (mosaic), blurring (Gaussian filters), black bars, kanting (outlining), and the use of digital avatars.

6.4 Deciding on an Approach

Sortinghat benutzt hashlib.sha1 Das ist ein Problem, da der algorithmus veraltet ist Sources: sortinghat repo <https://auth0.com/blog/sha-1-collision-attack/> <https://docs.python.org/3/library/hashlib.html#hashlib.sha1>

6.5 Implementing the Anonymization Server

6.5.1 Implementing the Server

6.5.2 Implementing the Pseudonymization

TODO: überschrift mit der eigentlichen Methode austauschen

6.5.3 Implementing the Anonymization

TODO: überschrift mit der eigentlichen Methode austauschen

6.5.4 Deploying the Server

Docker Container

7 Evaluation

lorem ipsum

8 Conclusions

8.1 Future Work

lorem ipsum

8. Conclusions

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